19 In experiment 1, a piece of calcium carbonate is placed in 0.200 mol dm^{-3} hydrochloric acid at 25 °C. The rate of the reaction is measured and recorded as rate 1.

In experiment 2, an identical piece of calcium carbonate is placed in 0.400 mol dm^{-3} hydrochloric acid at 25 °C. The rate of the reaction is measured and recorded as rate 2.

Which statement is correct?

A Rate 1 is greater than rate 2 because experiment 1 has a lower activation energy.

B Rate 1 is greater than rate 2 because experiment 1 has more collisions between reactant particles per second.

C Rate 1 is less than rate 2 because experiment 2 has a lower activation energy.

D Rate 1 is less than rate 2 because experiment 2 has more collisions between reactant particles per second.